

BEATWAVE: Contactless Drum Generation from Ambient Wireless Sensing

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Abstract—This work proposes BEATWAVE, a system that aims to enable musicians to “play the air” by transforming mid-air gestures into expressive drum beats using ambient wireless signals. Unlike conventional digital instruments that depend on touch surfaces, often cameras, or wearable sensors, BEATWAVE requires no dedicated devices. The system leverages motion-induced perturbations in pervasive wireless signals, extracting temporal features that are mapped to rhythmic events through a deep learning model. Through technical evaluation and user studies, we demonstrate that BEATWAVE is feasible, robust, and creatively engaging. Our results highlight the potential of ambient, device-free interfaces, offering new forms of embodied and accessible musical performance.

Index Terms—Device-free drumming, gesture assessment, HCI

I. INTRODUCTION

The evolution of musical interfaces has long shaped how humans create and experience sound. Acoustic instruments afford nuance and embodiment, while electronic synthesizers and drum machines offer versatility and control. Yet most digital instruments today still depend on physical surfaces, knobs, or wearables. Observing these, we raise a central question: *Can musical performance be made more free, expressive, and accessible without adding physical devices or sensors?* To address this question, we envision a musical system to technically “play the air”, as depicted in Fig. 1, where an individual has the ability to create musical notes in a device-free manner.

a) *Motivation*: Existing musical instruments, especially the digital instruments, are not always affordable, and their interfaces [1] each have certain limitations. For instance, Touch-based controllers [2] reduce rich gestures to taps, swipes, or button presses. Wearables [3] encumber performers and restrict freedom of motion. Therefore, a device-free interface removes barriers of cost, setup, and encumbrance while preserving spontaneity. We realised that there is a need to leverage a modality that captures spatio-temporal information while being device-free to best suit shaping this vision.

b) *Leveraging adequate modalities*: While several modalities have been explored for the study of instrumentless musicals, each has certain limitations. Vision-based ideas [4], [5] enable free-space control but are sensitive to lighting and occlusion. Also, their alignment and positioning play a crucial role. Inertial measurement units (IMU) [6]–[8] have been studied quite a few times in recent years to address

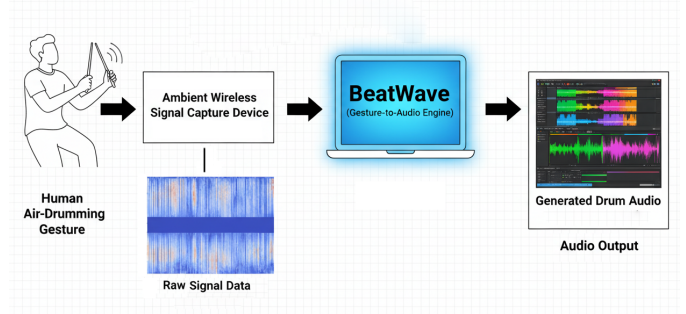


Fig. 1: The objective of BEATWAVE, where an individual’s mid-air playing of a drum is to be converted to beat audio.

the same problem. However, even though IMU can capture temporal information, the spatial information is limited. Brain-computer interfaces [9] offer fidelity, but their performance is yet to mature and remain impractical for casual use. However, wireless sensing poses an extremely potent solution to address the problem. The strong contenders include Ultra Wide Band (UWB), millimeter Wave (mmWave), and Wi-Fi Channel state Information (CSI) [10]. UWB [11], despite providing high spatial resolution, relies on specialized hardware and faces regulatory restrictions, limiting deployment. Millimeter wave (mmWave) offers fine-grained motion capture [12] due to its bandwidth advantage, but is highly sensitive to occlusions, susceptible to attenuation, and is often expensive when it comes to finding a device that matters. Therefore, using these options will automatically question the idea of a low-cost solution against the *costly musical instruments*. In contrast, CSI captures spatial information, device-free and sensitive to subtle movements [13]. It works robustly through occlusions, requires no instrumentation of the performer, and captures sufficient temporal structure to serve as a generative substrate for rhythmic interaction.

c) *Problem formulation*: In this work, we address the problem of generating rhythmic drum patterns from unconstrained hand gestures in free space. The core challenge lies in translating high-dimensional, temporally varying gesture-induced perturbations into discrete, beat-aligned drum events, preserving the performer’s intended rhythm and expressive timing. While the output is restricted to percussive beats, the problem remains inherently *generative*: the system must infer the temporal structure and velocity of gestures to produce

coherent, musically meaningful sequences. By this, we aim to open a pathway for natural, device-free percussion that emphasizes the tactile freedom of “playing the air.”

d) Contributions: That said, we introduce BEATWAVE, to fulfill these objectives. Our contributions in this regard are the following:

- 1) We design a device-free framework for a musical interface leveraging everyday wireless infrastructure to generate rhythmic drum beats, which, to the best of our knowledge, has never been attempted before.
- 2) We propose a temporal generative model, the core of BEATWAVE, to *generate* rhythmic sound patterns by mapping movement dynamics.
- 3) We curate a time series dataset of Drum audio, along with the corresponding CSI measurement.
- 4) We perform technical and user studies demonstrating the feasibility, usability, and creative potential of BEATWAVE

II. RELATED WORK

Progress in music interaction can broadly be understood along two dimensions: (i) the design of interfaces, and (ii) approaches for modeling music generation from gestures.

A. Interfaces and Interaction

Acoustic instruments [14] tightly couple gesture and sound with precision and expressivity. However, their high cost and limited accessibility constrain broader adoption. To address this, researchers have proposed alternative interfaces such as wearable-based interaction, accelerometer-driven systems, and glove-based haptic devices [15], [16]. While such approaches extend expressive possibilities, they inevitably reduce naturalness by requiring performers to carry additional hardware. More recently, modalities like inertial capture [8], marker tracking, and temporal motion encoding have been explored to enrich gesture recognition. Despite their promise, these techniques remain limited in capturing the full richness of spatial information during performance.

An alternative direction for gesture with relevant potential is *device-free wireless sensing*, which captures motion without cameras or wearables by analyzing perturbations in radio signals caused by body movement. For example, mmWave radars [12] and UWB systems [17] provide centimeter-level spatial resolution and precise range–Doppler profiles. However, they require dedicated transceivers and specialized antennas, are not very cost-effective, and are not as pervasive as Wifi. Wifi Channel State Information (CSI) offers a compelling contrast. CSI exposes fine-grained amplitude and phase responses across multiple subcarriers. Crucially, CSI can be extracted directly from off-the-shelf Wi-Fi devices, including those compliant with emerging standards such as Wi-Fi 6/7 and 5G NR [18]–[20]. This accessibility makes Wi-Fi CSI a practical foundation for scalable, device-free interaction.

B. Modeling of Gestures

Much of the existing work on gesture-driven interaction formulates the task either as a classification problem [17], [21]–[25] or as a localization problem [26]. There are a few works

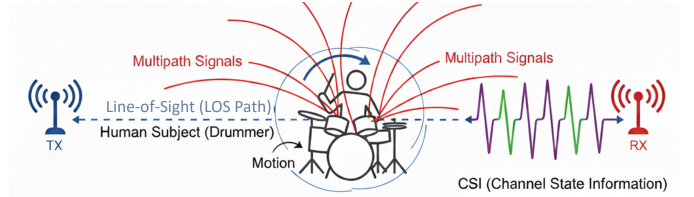


Fig. 2: Arm and hand movements modulate both line-of-sight and multipath components of the signal, producing amplitude and phase fluctuations observable across subcarriers.

that also traverse the path of regression [27]. While effective for recognition or tracking, these methods do not directly enable creative musical output. A generative perspective of gestures, on the other hand, remains relatively rare. A notable exception is [8], which explores using IMU signal dynamics to directly drive rhythm generation. However, such efforts are limited in scope, leaving unexplored the broader potential of generative mappings in device-free contexts. This gap points to an opportunity to reframe wireless sensing not merely as a recognition modality, but as a medium for creativity.

Our Position: Building on this gap, we propose BEATWAVE, which leverages Wi-Fi CSI to use mid-air gestures in generating rhythmic audio by treating wireless sensing simultaneously as an input modality and an expressive channel.

III. BACKGROUND AND FEASIBILITY STUDY

We now provide the fundamentals and our observations concerning the wireless interface.

A. CSI and Motion Sensitivity

CSI has been studied quite extensively [28]–[31] for a fine-grained characterization of different gesture activities in an environment, even in recent times. In orthogonal frequency-division multiplexing (OFDM) systems such as Wi-Fi, a transmitted signal has multiple subcarriers, each experiencing unique amplitude attenuation and phase shift.

Formally, the CSI for subcarrier k can be modeled as:

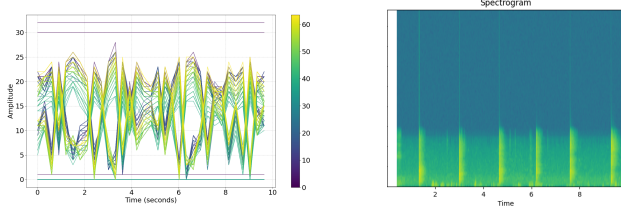
$$H_k = \sum_{p=1}^P \alpha_p e^{-j2\pi f_k \tau_p},$$

where α_p and τ_p represent the complex attenuation and delay of the p -th multipath component, and f_k is the frequency of subcarrier k . In practice, CSI is expressed in polar form as:

$$H_k = |H_k| e^{j\phi_k},$$

where $|H_k|$ captures amplitude attenuation and ϕ_k reflects phase shifts.

Human motion perturbs both α_p and τ_p , thereby inducing temporal variations in $|H_k|$ and ϕ_k . When a drummer makes rhythmic gestures, such as arm swings or hand strikes, these perturbations exhibit periodicity across subcarriers, embedding a motion-dependent signature in the CSI stream. Fig. 2 illustrates how drumming motions disrupt line-of-sight and multipath propagation, yielding structured fluctuations in CSI.



(a) Subcarrier amplitude response. (b) Relevant audio spectrogram.

Fig. 3: Observation of CSI amplitude dynamics (a) and the audio spectrogram (b) during rhythmic drumming.

B. Feasibility Study

Before designing a solution, we first investigated whether drumming gestures leave identifiable rhythmic traces in CSI. The objective was not to evaluate end-to-end performance, but to establish that embodied percussion produces reliable, structured CSI dynamics aligned with beats.

a) Setup: For validation, a participant was positioned between a commodity Wi-Fi transmitter and receiver while performing drumming or clapping. We simultaneously recorded: (1) CSI across multiple subcarriers, and (2) audio to capture beat timing. All signals were time-synchronized for analysis.

b) Observation One: A visual analysis of the recorded signals, presented in Fig. 3, reveals pronounced and periodic fluctuations within the CSI magnitude traces. These results validate CSI as a generative substrate for mapping free-space motion into rhythmic outputs, establishing our core notion.

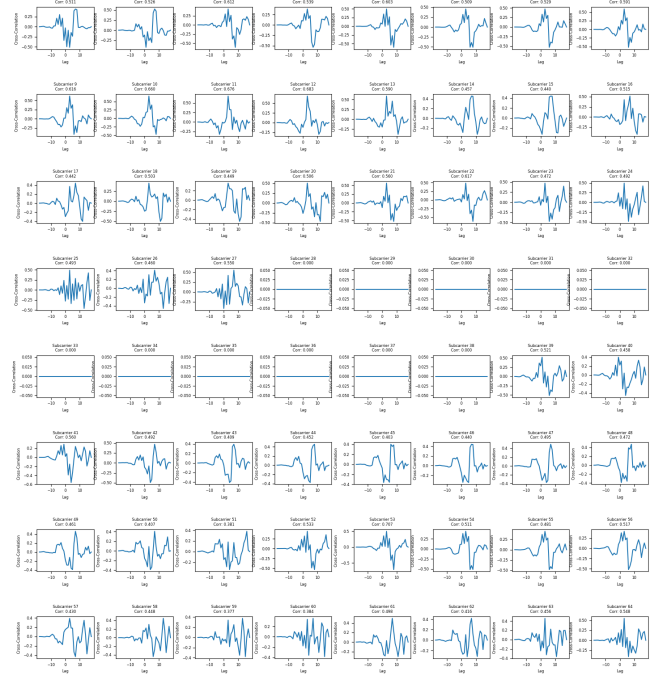
c) Observation Two: As expected, and as CSI behaviour was explored by earlier studies [32], although some channels are noisy, several exhibit a non-zero alignment ($r > 0. \times \times$) with audio events. The behaviour assessed for a 64-subcarrier transmission, by our CSI measurement device, is shown in Fig. 4. On exploring the cross-correlation of different subcarriers, we find that only some are actually responding. Interestingly, for the system to be deployable across different commodity setups, the correlated subcarriers cannot be known a priori and should be realised dynamically.

d) Observation Three: We also observe that calculating the correlation at different time lags is crucial because it helps us discover *cause-and-effect relationships* that aren't perfectly simultaneous. For example, a person's movement might disturb the Wi-Fi signal (CSI change) a few milliseconds *before* they make an audible sound. The time lag helps find the delay where the two signals are most strongly related.

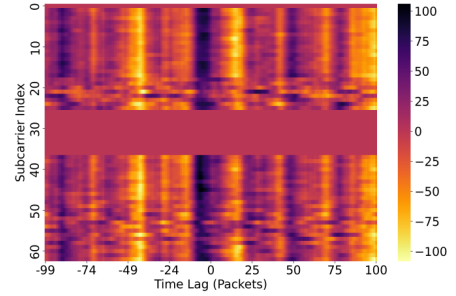
Therefore, shaping BEATWAVE requires suitable subcarriers that demonstrate adequate dynamicity, have a strong correlation with percussion dynamics, and also address the time lag.

IV. BEATWAVE

This section delineates the methodology of our proposed solution, BEATWAVE. The complete processing pipeline is stated in Fig. 5. We detail the pre-processing, the architectures of the model involved, and the comprehensive training below.



(a) Individual subcarrier response



(b) Summarized subcarrier response

Fig. 4: Correlation between CSI subcarriers' response to hand movements and audio beat timings (a) is further summarized as a heatmap and shown in (b). The darker color signify a higher correlation.

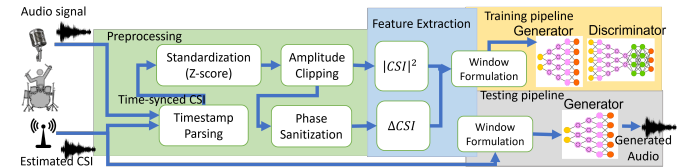


Fig. 5: Shows the complete process of BEATWAVE

A. Data Pre-Processing

The initial phase of our system involves processing raw CSI and audio data to create a dataset suitable for training the generative model. It involves a few pre-processing steps.

Data Structuring: The raw data is expected to consist of CSI logs and corresponding audio recordings as sensed by the respective instruments. Both are timestamped, thereby

facilitating time synchronization. We establish a one-to-one correspondence between the data pairs by parsing timestamps embedded within each subsequent captured data point. Once paired, the data becomes a time-synced series of CSI and corresponding audio samples. However, their measurement frequency differs.

Lightweight CSI Feature Engineering: To maintain computational efficiency, we employ a lightweight feature extraction process. We employ a variance threshold to identify the most suited set of 8 CSI subcarrier channels. This approach is advantageous for two primary reasons: 1) It automatically filters out unresponsive or “null” subcarriers (i.e., those exhibiting zero or near-zero variance). 2) It serves as a feature selection mechanism, allowing us to isolate channels with the highest variance. Raw CSI data is susceptible to environmental noise and hardware-induced offsets. To create a robust input for our model, we applied a multi-stage pipeline further stated as follows:

Filtering: Raw CSI signals have noise that can mask the subtle modulations induced by human motion. A low-pass Butterworth filter [33] is applied to the raw signal to remove high-frequency noise, and a Hampel filter [34] is used to eliminate sporadic outliers in the CSI amplitude streams.

Phase Sanitization: We corrected for random phase offsets by applying a linear transformation to the raw phase data, which produced a stable and continuous phase response.

Feature Extraction: From the cleaned CSI windows, we extracted a set of features sensitive to motion, including: energy variations, $|CSI|^2$ (squared absolute value of CSI), which captures the signal power fluctuations. First-Order Differences, which is the difference between consecutive CSI values (ΔCSI), highlighting the rate of change in the channel. First-order amplitude changes, and the moving variance of both amplitude and phase. This results in a feature tensor of shape $(8 \times 2 \times 2048)$ for each data chunk, which is then flattened into (16×2048) before being fed into the model.

B. Model Architecture

The core of BEATWAVE is based on a Conditional Generative Adversarial Network (cGAN) [35]. The overall architecture is depicted in Fig. 6.

Generator: The generator’s role is to synthesize a raw audio waveform from a latent noise vector and a corresponding CSI feature chunk. Its architecture is as follows:

Input Layer: The input is a concatenation of a latent vector of dimension 48 and the flattened CSI feature tensor.

Fully Connected Layers: A series of dense layers (*nn.Linear*) project the high-dimensional input into a suitable shape for the convolutional layers, progressing from the total input dimension to 256, then 128, and finally to a tensor of size 64×32 . *Leaky ReLU* activation and *Dropout* are used for non-linearity and regularization.

Transposed Convolutional Layers: The core of the generator consists of a stack of 1D transposed convolutional layers (*nn.ConvTranspose1d*) that progressively upsample the feature maps through channel sequences of $64 \rightarrow 48 \rightarrow 32 \rightarrow 16 \rightarrow$

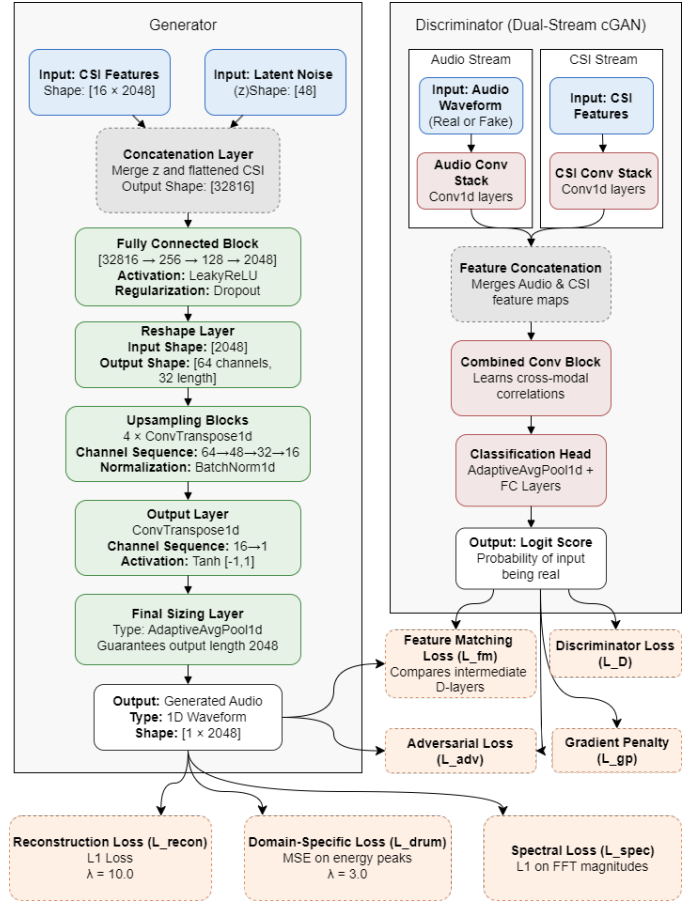


Fig. 6: The proposed dual-stream cGAN architecture. The generator synthesizes audio from CSI and latent noise. The discriminator receives both audio and CSI streams to differentiate between real and generated pairs.

1. *BatchNorm1d* is applied after each layer to stabilize training. **Output Layer:** A final transposed convolutional layer maps the features to a single audio channel, followed by a *Tanh* activation function to ensure the output waveform is bounded within $[-1, 1]$. An *AdaptiveAvgPool1d* layer guarantees the output sequence length matches the target chunk size of 2048.

Discriminator: The discriminator is implemented as a dual-stream deep convolutional network designed to differentiate between real and generated audio samples, conditioned on the associated CSI data. This architecture allows the model to learn features from each modality independently before combining them:

Dual Input Streams: The model processes two inputs in parallel: the audio waveform (1 channel) and the CSI feature tensor (16 channels - $|CSI|^2$ and ΔCSI).

Parallel Convolutions: Each input stream is passed through its own dedicated stack of 1D convolutional layers (*nn.Conv1d*). These layers are enhanced with *Spectral Normalization* (*spectral_norm*) to enforce the 1-Lipschitz constraint, a key technique for stabilizing GAN training.

Feature Concatenation: The feature maps produced by the two parallel streams are concatenated along the channel dimension

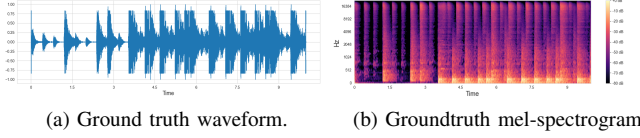


Fig. 7: An example of a ground truth audio sample from the dataset, shown in the time domain (waveform) and frequency domain (mel spectrogram). BEATWAVE’s objective is to replicate these complex structures.

into a single tensor of 256 channels.

Combined Processing: The merged feature map is processed by a final convolutional block to learn correlations between the audio and CSI features.

Classification Head: An *AdaptiveAvgPool1d* layer flattens the final feature map, which is then passed through fully connected layers to produce a single logit score, representing the probability of the input audio being real.

C. Training and Evaluation process

The models are trained end-to-end using an adversarial process guided by a composite loss function. The training process is optimized using the *AdamW* optimizer and an *ExponentialLR* scheduler, which activates after a set number of warmup epochs. In the training loop, for every generator update, the discriminator is updated n_critic (2) times [36].

Generator Loss: The generator is optimized using a multi-component loss function, L_G , designed to ensure both realism and fidelity:

$$L_G = L_{adv} + \lambda_{recon}L_{recon} + \lambda_{fm}L_{fm} + \lambda_{spec}L_{spec} + \lambda_{drum}L_{drum}$$

where:

- L_{adv} is the adversarial loss (BCEWithLogitsLoss).
- L_{recon} is an L1 reconstruction loss weighted by $\lambda_{recon} = 10.0$.
- L_{fm} is a feature matching loss weighted by $\lambda_{fm} = 2.0$.
- L_{spec} is a spectral loss weighted by $\lambda_{spec} = 1.0$.
- L_{drum} is MSE loss weighted by $\lambda_{drum} = 3.0$.

Discriminator Loss: The discriminator is optimized using a standard GAN objective stabilized with a gradient penalty. Its loss function, L_D , is:

$$L_D = L_{real} + L_{fake} + \lambda_{gp}L_{gp}$$

where L_{real} and L_{fake} are the binary cross-entropy losses for real and fake batches, respectively, and L_{gp} is the gradient penalty, weighted by $\lambda_{gp} = 5.0$.

V. EXPERIMENTAL SETUP AND DATASET

To study and evaluate the BEATWAVE, we curate a custom dataset by recording time-stamped CSI and audio from live drumming performances. The development of this dataset was a foundational step, as no public equivalent exists for this specific generative task.

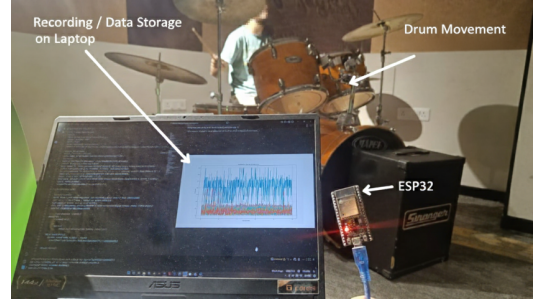


Fig. 8: Hardware setup and environment.

A. Data Collection

Hardware and Environment: Our data collection setup utilized a single ESP32 microcontroller for self-contained CSI sensing. The initial preparation involved setting up the ESP-IDF v4.3 [37] development environment on a host computer. We then cloned the ESP32-CSI-Tool repository [38], which provided the firmware base. Using the tool, we configured the Wifi credentials, enabled Wi-Fi CSI component, and setting a high serial baud rate (e.g., 921600) to ensure high-throughput data transmission without lag. This customized firmware was then compiled and flashed onto the ESP32 device.

For data collection, the programmed ESP32 was positioned in a music room near an acoustic drum kit, along with a wifi transmitting source device, to best capture the signal modulations caused by the drummer’s movements. A scenario of the setup is shown in Fig. 8. CSI data was streamed from the device’s serial port to the host computer. To facilitate synchronization with other data sources, a Python script was used to append a host-based timestamp to each incoming CSI packet before logging it to a .csv file. Concurrently, audio was recorded on the same host device using a microphone to capture a high-fidelity ground truth of the drum sounds. This synchronized dataset was subsequently partitioned into training and testing sets to develop and evaluate our model with a split of 80-20.

Participants and Procedure: Data was collected from 4 volunteers (ages 20-24) with a range of drumming experience from novice to intermediate. To capture naturalistic performance data, drummers were not prompted with repetitive tasks. Instead, they were asked to play a variety of common grooves and rhythms in continuous sessions. This method produced a combined dataset of about 1.5 hours of audio-CSI streams, rich with the subtle variations inherent in genuine drumming.

B. Annotation

The recorded audio of the drum performances served as the ground truth for the CSI data. We processed the audio using an onset detection algorithm, specifically *librosa.onset.onset_detect*, to automatically extract the precise timestamps of each drum strike. These timestamps provided the exact moment a percussive event occurred. The continuous, preprocessed CSI data stream was then segmented into windows centered around these ground-truth timestamps.

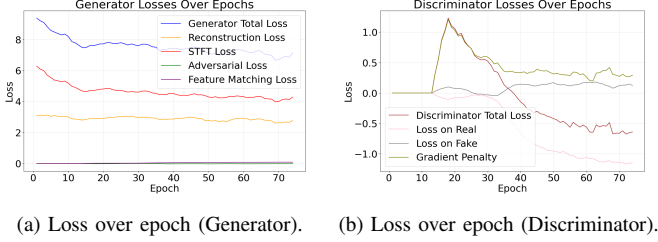


Fig. 9: Training metrics - Loss over epochs.

VI. RESULTS AND DISCUSSION

In this section, we present a comprehensive analysis of BEATWAVE’s performance. We evaluate the results both quantitatively, by examining the training metrics, and qualitatively, by inspecting the generated audio waveforms and spectrograms. We conclude with a discussion of the findings and potential limitations of our approach.

A. Quantitative Performance Analysis

The stability of the training process and the model’s convergence are best understood by analyzing the loss functions over time, as depicted in Fig. 9. The figure illustrates the behavior of the multi-component loss functions for both the generator and the discriminator. The curves demonstrate a stable adversarial dynamic; neither the generator nor the discriminator loss diverges, indicating that the model avoids common training pitfalls such as mode collapse. The symmetrical logarithmic scale reveals that while the total losses fluctuate, they remain within a bounded range, suggesting a successful convergence.

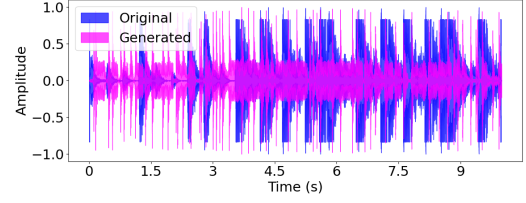
B. Qualitative Analysis of Generated Audio

Beyond numerical metrics, Fig. 10 provides a side-by-side comparison of a ground truth audio sample and a sample generated by BEATWAVE from its corresponding CSI data, evaluated in both the time and frequency domains.

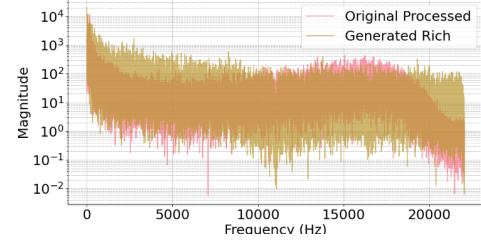
The time-domain comparison, shown in Fig. 10(a), illustrates BEATWAVE’s proficiency in capturing the temporal dynamics of the audio. The generated waveform successfully replicates the ground truth’s overall amplitude envelope, indicating that BEATWAVE could learned to map CSI fluctuations to the rhythmic structure of the sound. In the frequency domain, the frequency spectrogram comparison in Fig. 10(b) provides insight into BEATWAVE’s ability to reconstruct the spectral fingerprint of the audio. The generated spectrogram successfully replicates the fundamental frequencies, preserving the energy distribution across components. This indicates that BEATWAVE learns the essential “tone color” characteristics of the sound.

C. Audio Feature Similarity Analysis

To provide a more objective measure of quality beyond visual inspection, we analyzed the structural similarity of key audio features between the original and generated samples. Fig. 11 shows the distribution of cosine similarity scores for a



(a) Waveform comparison (Ground Truth vs. Generated)



(b) Frequency spectrogram comparison (Ground Truth vs. Generated)

Fig. 10: Qualitative comparison of a generated audio sample against its ground truth. The model effectively reconstructs the overall amplitude envelope and the dominant frequencies.

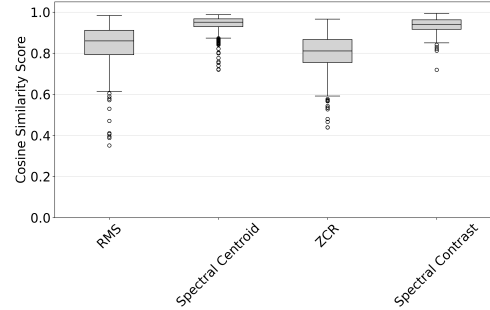


Fig. 11: Plot showing the distribution of cosine similarity between audio features of the original and generated samples.

set of foundational audio features. This metric evaluates how well the patterns and contours of each feature are preserved in the generated audio.

The high median similarity scores across all evaluated features are particularly encouraging. Strong performance on complex, multi-dimensional features like Spectral Contrast indicates that our model effectively preserves the spectral shape of the source audio. Similarly, high similarity in lower-level features like RMS Energy and Zero-Crossing Rate (ZCR) suggests that the model accurately reproduces the large-scale behavior of loudness and the general texture of the sound. This feature-level similarity provides quantitative evidence that the generated audio is not just superficially similar but structurally congruent with the original.

D. User Perception assessment

We also conducted a user perception survey to evaluate how humans experience and interpret the generated audio. For this

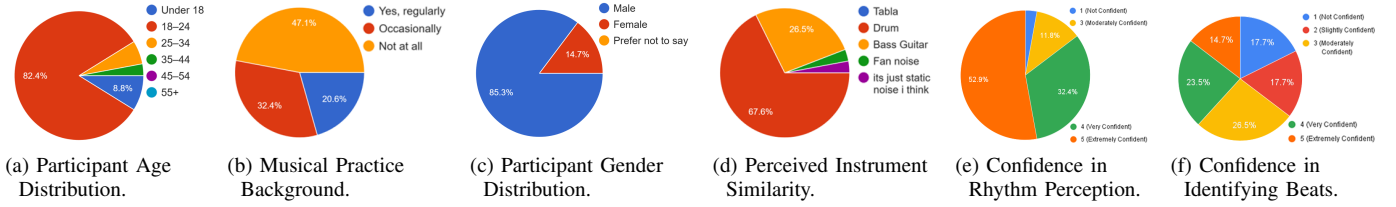


Fig. 12: The results from the user perception survey showing demographics and user views.

survey, we shared a few samples of the best of our generated audio samples to people across different geographic locations and varying demographics without disclosing any information about the objective of the work nor what the audio samples are about. The survey collected responses on the *identification of the instrument, the rhythmic quality of the audio, and the listener’s confidence on perceived nature of the sound*, thus signifying the quality. Till the writing of the manuscript, we received views from about 35 participants, summarized in Fig. 12, and are as follows:

The participant demographic was predominantly young adults, with 82.4% of respondents in the 18–24 age group (Fig. 12(a)). Regarding musical experience, the group was varied: 20.6% practice regularly, 32.4% play occasionally, and 47.1% do not practice music at all (Fig. 12(b)), suggesting the findings do include a significant number of musically trained listeners whose views are expected to be considered with a higher degree of confidence.

A key finding was the consistent identification of the sound’s character. An overwhelming majority (67.6%) perceived the generated audio as being most similar to a drum, with another 26.5% likening it to a bass guitar (Fig. 12(d)). This indicates that the model is highly effective at synthesizing low-frequency, percussive sounds that form the rhythmic foundation of music.

This was further corroborated when participants were asked about rhythmic structure. A combined 85.3% of respondents were either ‘Very Confident’ (32.4%) or ‘Extremely Confident’ (52.9%) that the sounds formed a proper rhythmic pattern or beats (Fig 12(e)). This provides strong evidence that the model successfully translates the temporal fluctuations in CSI data into a coherent and recognizable rhythm.

When asked about their confidence in identifying beats solely through the generated audio, responses were more varied (Fig. 12(f)). While a plurality of users expressed moderate to high confidence, the distribution suggests that while the macro-rhythmic structure is clear, but not the finer details, aligning with our earlier observation that the model may smooth over some high-frequency textural information. The Mean Opinion Score (MOS) for confidence in identifying beats is 3.03, which is quite encouraging. The 95% confidence interval for this score is between 2.58 and 3.48.

E. Discussion

The results highlight several important insights. First, the training dynamics indicate a stable adversarial process where

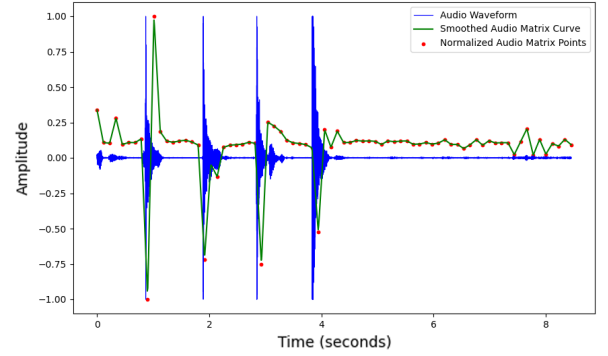


Fig. 13: Audio signal normalization using CSI amplitude.

generator and discriminator losses remain bounded, confirming that the model avoids pitfalls such as mode collapse and achieves consistent convergence. Second, waveform and spectrogram comparisons show that the model not only captures the temporal rhythm but also reproduces the essential spectral fingerprint of percussive sounds, validating its ability to preserve both macro-level rhythmic envelopes and micro-level tonal color. Third, feature-level similarity analysis reinforces this by demonstrating high congruence across diverse descriptors ranging from RMS Energy to Spectral Contrast, suggesting that the generated audio is structurally faithful rather than superficially imitative. Fourth, the user study provides external validation: listeners consistently perceived the audio as drum-like and rhythmically coherent, indicating that the system generalizes well to human perception beyond quantitative similarity. Finally, the varied confidence in beat identification reveals a nuanced limitation. While the system captures rhythm and low-frequency percussive qualities effectively, it tends to smooth out fine-grained high-frequency details, highlighting a direction for future refinement.

VII. ABLATION STUDY

To better understand the impact of individual design decisions in BEATWAVE, we performed an ablation study focusing on three aspects: (i) audio normalization strategies, (ii) loss function choice, and (iii) the effect of temporal window size.

A. Audio Signal Normalization Strategy

The initial approach involved pre-normalizing the ground-truth audio waveform based on the corresponding CSI amplitude fluctuations. This assumes a simplistic, linear corre-

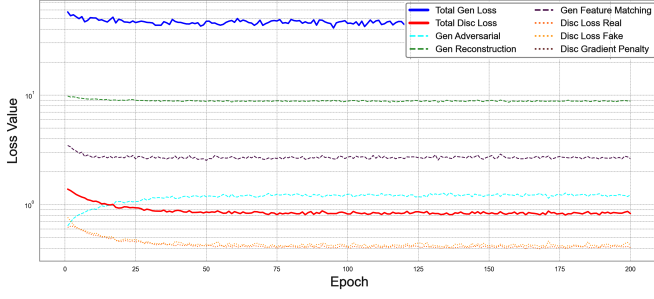


Fig. 14: Training metrics for the GAN using the Binary Cross-Entropy (BCE) loss function.

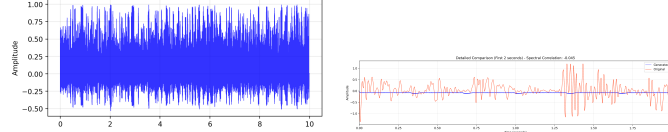


Fig. 15: Comparison of generated audio quality based on the input data window size. The shorter 2-second window (b) produces a more rhythmic output than the full 10-second window (a).

lation between the two modalities. However, this approach was rejected as it invariably leads to a significant loss of information and distortion of the ground-truth audio as in Fig. 13. Instead of imposing a flawed prior, the ground-truth audio is independently normalized to a $[-1, 1]$ range to match the generator’s *nn.Tanh* output, providing a stable target. The chosen Wasserstein GAN with Gradient Penalty (WGAN-GP) architecture is robust enough to discover the intricate correlations between the processed CSI features and the normalized audio without requiring destructive pre-processing of the audio signal itself. The network, therefore, learns the appropriate “smoothing” and scaling as part of its training.

B. Loss Functions

We explored multiple Loss functions in the study. On the use of Binary Cross-Entropy (BCE) Loss, the training objective is a minimax game. The discriminator is trained to minimize the loss for classifying real data as real, and fake data as fake. The generator is trained to maximize the discriminator’s confusion. This approach, however, frequently suffers from vanishing gradients and mode collapse, as observable in Fig. 14.

With the *Wasserstein Loss with Gradient Penalty (WGAN-GP)*, ensure that the discriminator is repurposed as a “critic” that outputs an unbounded score of realness rather than a probability. The critic’s loss is the Wasserstein distance, which it tries to maximize. The generator’s objective is to minimize the critic’s score for its fake samples. The key to its stability is the Gradient Penalty (GP), added to the critic’s loss. It enforces the 1-Lipschitz constraint by penalizing the critic if the norm of its gradient compared to its input deviates from 1. The observable results were already shown in Fig. 9.

C. Time Window Consideration

Processing full-length, high-sample-rate audio and CSI streams simultaneously is computationally prohibitive and leads to less accurate or incoherent results. This can be observed in Fig. 15(a) where the whole 10-second audio and CSI data streams were fed to the model, and in Fig. 15(b) where 2-second windows were provided. The full-sequence input results in a high-entropy, noise-like output, indicating a failure to learn meaningful long-range dependencies. In contrast, the shorter window enables the model to capture localized temporal features and converge on the underlying rhythmic structure of the data. This observation suggests that shorter segment lengths are critical for learning the high-frequency patterns that define percussive audio. Therefore, BEATWAVE employs a “chunking” methodology, segmenting the data streams into shorter, 2048-sample windows.

VIII. CONCLUSION AND FUTURE DIRECTIONS

BEATWAVE proposes a promising contribution to the ever-growing pursuit of finding intelligent musical interfaces [39]. Exploiting the ambient wireless (Wifi) signal, BEATWAVE generates drum beats in a device-free manner, opening a new direction for interactive music composition. More interestingly, BEATWAVE addresses the problem of music generation in a *generative* manner rather than a classification [40] or regression-based [27] idea. To our findings, BEATWAVE, could preserve the overall amplitude envelope of the generated signal and could preserve the dominant frequency instances. However, we did observe a fair amount of loss in the detail of the generated sounds, which is quite reflected in our survey outcomes when assessed by human participants. Overall, we believe, BEATWAVE could open a new direction for device-free musical interface.

Even though the current processing pipeline is designed for real-time audio generation, the idea has not been tested in a real-time manner and remains as future work. Additionally, the setup for data collection and evaluation was pretty much controlled with the least possible unwanted dynamics, thus avoiding noise. In the future, we would like to address external noise [41] due to unwanted movements. We would also like to explore high bandwidth alternatives such as Wi-Fi 6 and Wi-Fi 7 for a finer capturing of musical activities. Future work will also focus on model architectural enhancements, such as incorporating attention mechanisms, to better capture these fine-grained details. Additionally, expanding the training dataset to include a wider variety of acoustic events could further improve BEATWAVE’s generalisation capabilities. The source code used in this study is available in our online repository ¹. to support reproducibility.

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¹<https://anonymous.4open.science/r/ipynb-375A>

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